

8th CII 3M Compétition 2020

A Kaizen Presentation By

(Everest Industries Ltd)

Podanur Works - Coimbatore

18th September 2020

COMPANY OVERVIEW

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

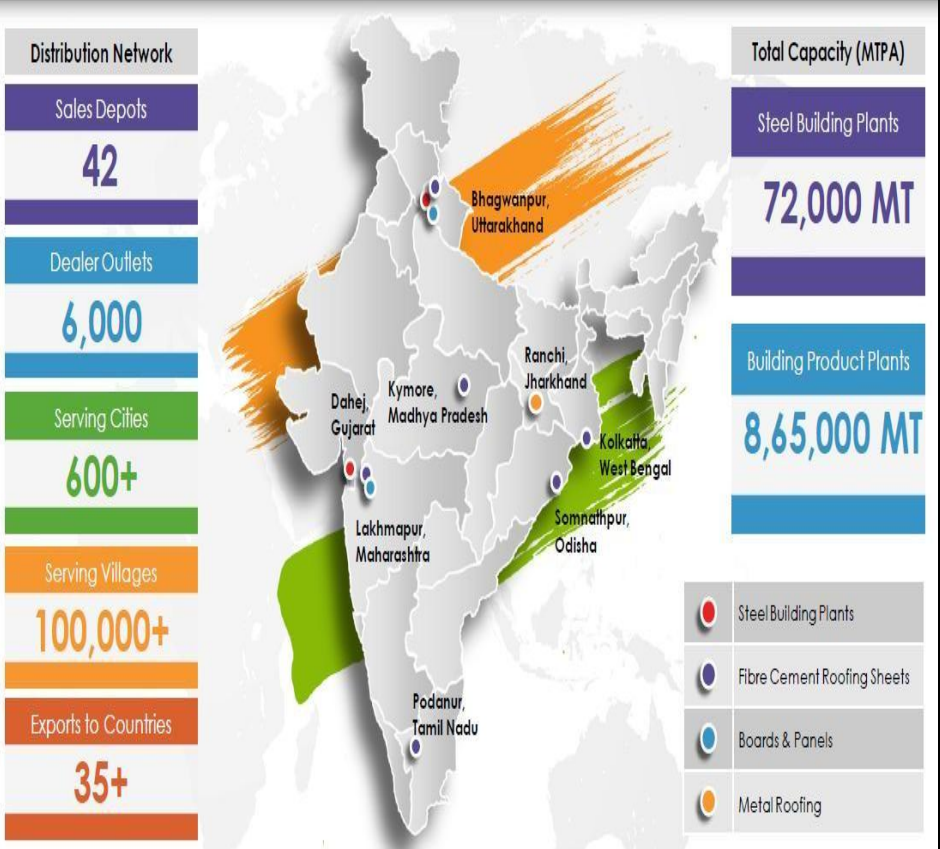
OUR BUSINESS

The company businesses are divided into two distinct business segments with capacity –

- 1. Buildings Products – 8,65,000 MT / Annum
- 2. Steel Buildings – 72,000 MT / Annum

Buildings Products segment contributed 63% of the company’s revenue and includes Fiber cement roofing sheets, Fiber cement boards, Rapicon wall & their accessories.

Steel Building segment accounted for 37% of annual revenue. In this segment Company offers customized building solutions in steel such as Pre-Engineered steel Buildings, Smart steel buildings, Metal roofing and Cladding.



Products:



Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Customers:



Kaizen Theme: To eliminate green sheet wastage.

Implemented Area: Line No -1, Podanur Works, Coimbatore

Implementation Date: 13th November 2019

Cross Functional Team:



Mr. Y. Kesavaraj
(Prod.Line Incharge)



Mr. M. Vijay Kumar
(Electrical Department)



Mr. N. Karthikeyan
(Prod.Shift Incharge)



Mr. N. Anil Kumar
(Engineer - QS & BE)



Mr. Pillai
(Fabricator)



Mr. Narayanasami
(Board Operator)

Theme

To eliminate Green sheet wastage.

Reason for selecting Theme :

- *Sending green sheet to recycling unit leads to production loss.*
- *Recycling & re-producing sheets increases power consumption.*
- *Recycling & re-producing sheets increases conversion cost.*

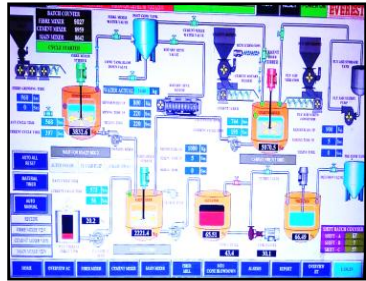
Problem Status :

- Loss of productivity to recycling unit during De-piling bogie changeover.

Sheet formation from Hatschek machine is continuous un-interrupted process. So, we have to pile the sheets in the bogie along with templates immediately in piling M/C. If sheet piling is delayed, another sheets which will be produced by machine will overlap the sheet waiting in piling M/C. By this both 2 sheets will be wasted for piling. So, if there is delay in piling, we are sending 1 sheet to recycling unit.

Problem Description

Process Flow:

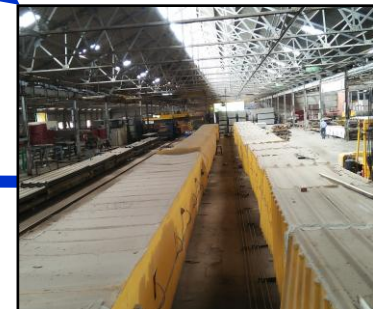


1) RAW MATERIAL
PREPARATION

2) SHEET FORMING

3) SHEET HANDLING/
PIILING MACHINE

SHEET RECYCLING UNIT



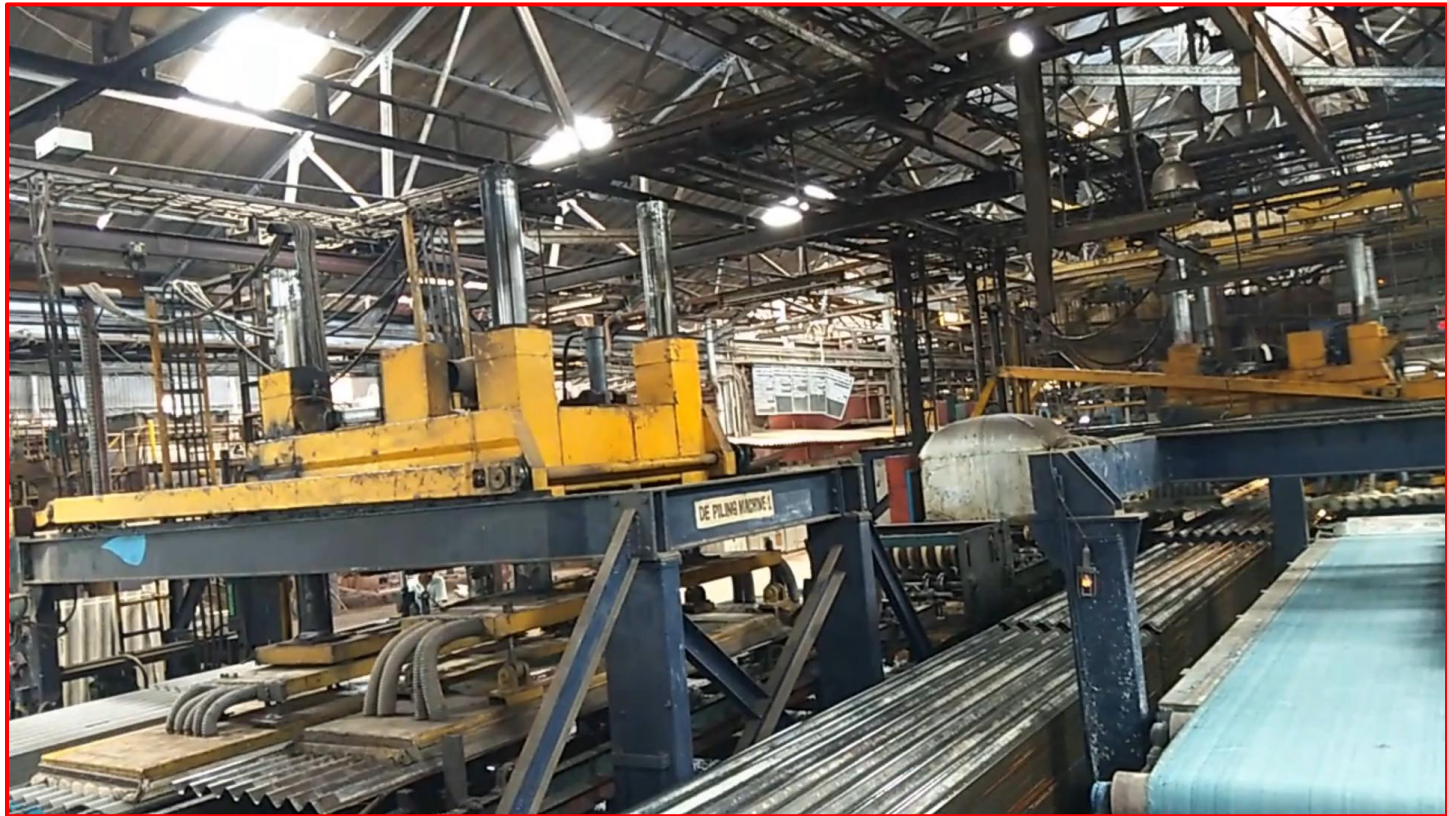
6) TEMPLATE OILING
SYSTEM

5) DE-PIILING MACHINE

4) PRE-CURING TUNNEL

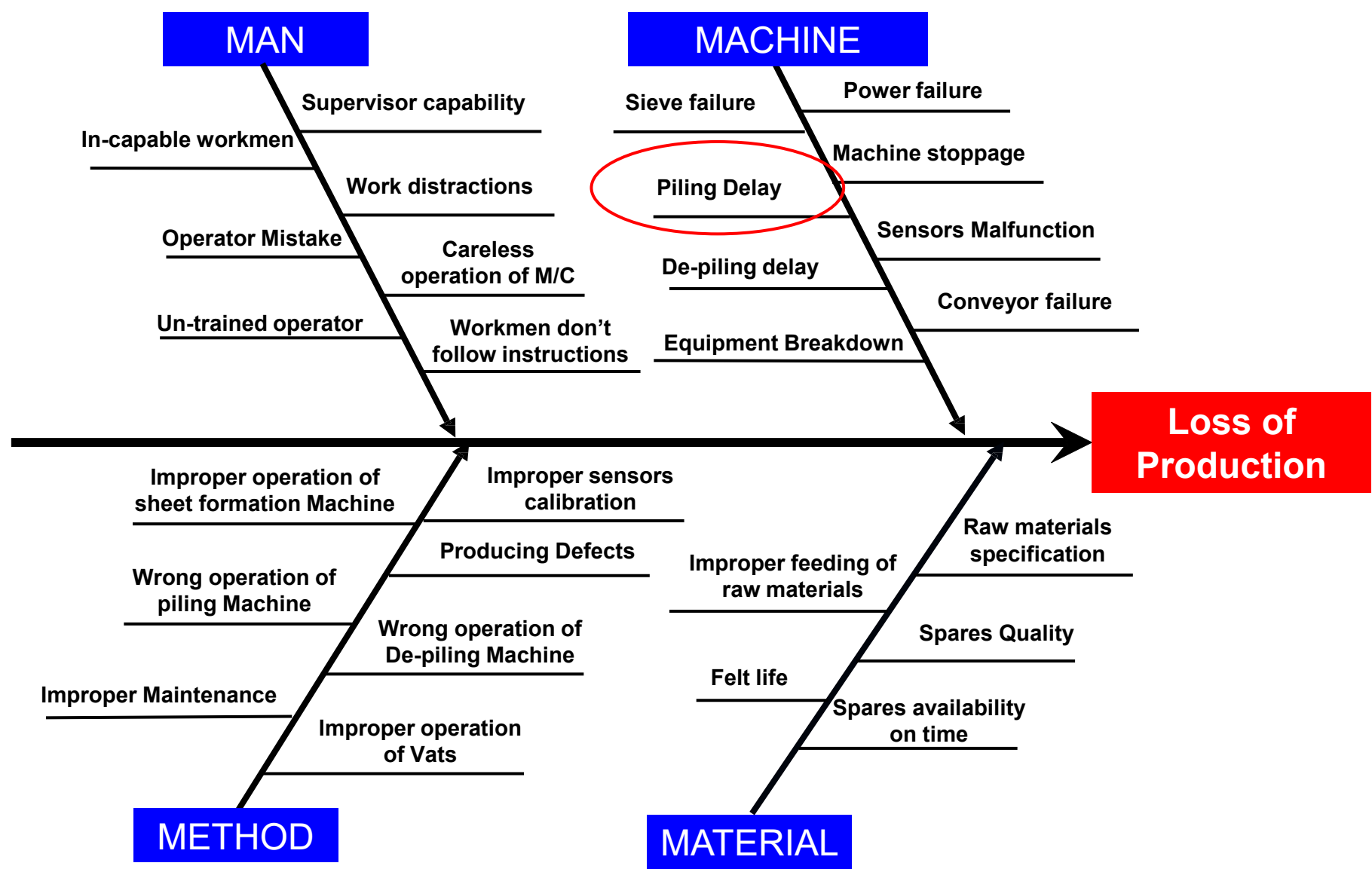
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Cause & Effect Diagram (Analysis)



Problem: Loss of production at Piling station.

Why 1

Green Sheet are sent to recycling unit.

Why 2

Green sheets can't be piled in the bogie by piling M/C.

Why 3

Due to template un-availability.

Why 4

There are no templates available to supply, for piling Green sheets in bogie.

Root Cause:

There are no templates available to supply, for piling Green sheets in bogie.

Idea Generation: To provide extra conveyor for storing the templates & to supply the stored template to piling machine during template unavailability.



Steps involved in kaizen development					
S.No	Steps	Start Date	Target Date	Responsibility	Status
01	To arrange space for extending the template roller conveyor system.	16/10/2019	17/10/2019	Production Department	Completed
02	To install roller conveyor frame.	18/10/2019	31/10/2019	Maintenance Department	Completed
03	To install rollers, Gearbox, Motors & to couple them with chain sprockets.	01/11/2019	10/11/2019	Maintenance Department	Completed
04	Giving control circuit at De-piling Board & taken safety measures.	11/11/2019	13/11/2019	Electrical Department	Completed



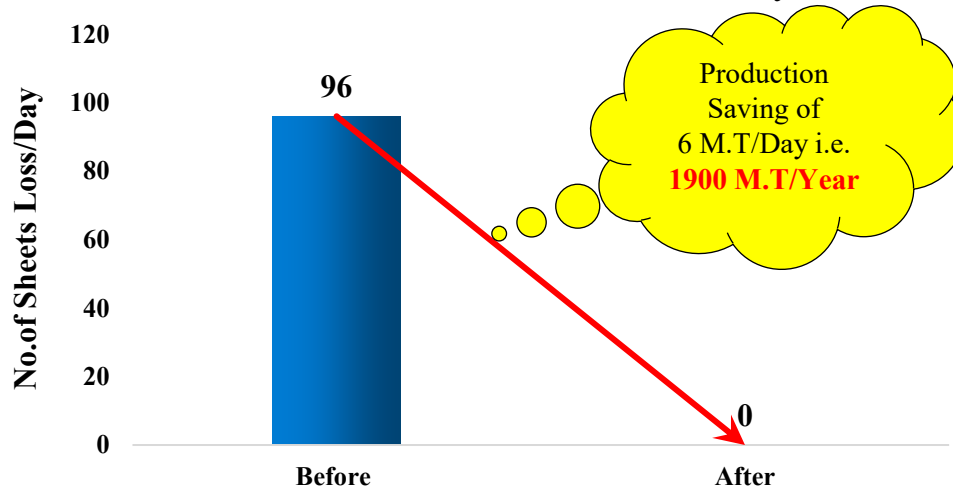


- ❖ Now, during un-availability of templates for piling sheets, templates are supplied from additional template storage conveyor. By this we are saving 1 sheet/De-piling bogie changeover.

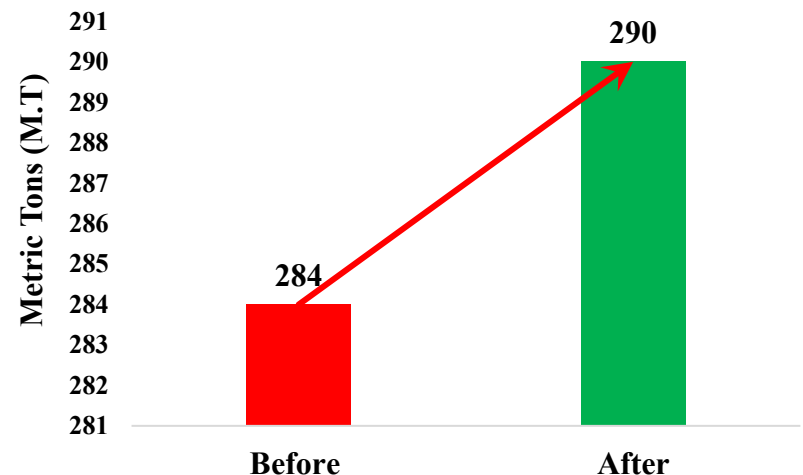
Productivity Improvement by eliminating Muda

	Before	After
Green sheet loss per bogie changeover	1 sheet/Bogie	0 sheet/Bogie
Total Bogie changeovers in a Day	96	96
Total Sheet loss to recycling unit / Day	96 sheets	0 sheets
Loss of production loss M.T / Day	6 M.T	0 M.T
Production Saving by eliminating Muda	158 M.T/Month	
Productivity Saving / Year	1900 M.T / Year	

Green Sheets Production Loss / Day



Average Monthly Production



Benefits:

- 1) **Productivity** - Productivity improvement of 1900 M.T/Year.
- 2) **Quality** - Quality improved due to reduction in consumption of recycled waste.
- 3) **Cost** - Cost Saving of INR. 1.36 Cr/Annum with less investment Cost of INR.20,000/- due to in-house built.
- 4) **Delivery** - Machine production capacity has increased by 1.85%.
- 5) **Safety** - No Safety issues.
- 6) **Morale** - Morale increased due to productivity improvement.

Validation of Kaizen in 7 Wastes of Manufacturing		
T	Transportation	Transportation of sheet to recycling unit eliminated.
I	Inventory	NA
M	Motion	NA
W	Waiting	Piling M/C waiting due to template un-availability eliminated
O	Over-Production	Market Demand = Production Capacity
O	Over-processing	Producing sheet by consuming recycling material is reduced.
D	Defects	Quality improvement due to less recycled waste consumption.

- Training on Updated Work Instructions are given to all the workmen for Operating, regular inspection & maintenance.

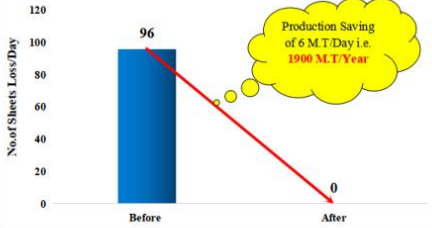
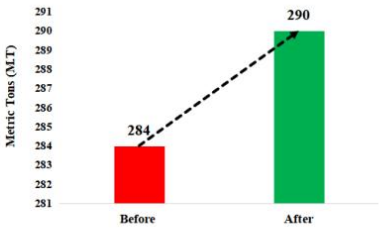
	EVEREST INDUSTRIES LIMITED PODANUR WORKS IS/ISO 9001 : 2015 WORK INSTRUCTION	DOC NO : EL/WI/513 REV NO : 01 PAGE : 1 OF 1 DATE : 15/11/2019
	Work Instruction: WI 513 DE-PIILING OPERATION: Procedure: 1. Wait till the filled(De-piling) bogie take position in the De-piling machine & raise the bogie, till it reaches required height. 2. Separate templates from the bogie by using template separation pad & separate sheet from the bogie with sheet separation pad. 3. Place separated templates on the roller conveyor, to pass templates to piling machine through template oiling unit. 4. Stack separated sheets on the sheet trolley. 5. When Piling bogie changeover takes place, store one template on additional roller conveyor by taking reverse. 6. When De-piling bogie is empty, bring the empty bogie down & change the bogie. 7. In the mean time, supply template from the additional roller conveyor to piling machine. So that, template supply to piling machine is un-interrupted & piling machine won't stop piling sheets due to template unavailability.	

Kaizen Sustenance:

WHAT TO DO	Regular Inspection & Maintenance.
HOW TO DO	Regular Training on Standardized Work instructions
FREQUENCY	Weekly

Horizontal Deployment:

Line No.1	Line No. 2
Implemented	Implemented

 Confederation of Indian Industry  INSTITUTE of QUALITY	8th Edition of 3M Competition 3M KAIZEN IDEA Muda/Mura/Muri Tick wherever necessary	Activity (Please select any one category only)	Muda ☒	Mura ☐	Muri ☐	 Everest Industries Limited										
Plant : Podanur, Coimbatore Kaizen Theme : To eliminate Green sheets wastage.	Area/Line/Machine : Shop Floor Idea (Logical correlation with root cause) : To install additional roller conveyor to store extra templates & supply them for piling sheets in bogie when templates are not available .															
Problem/Present Status : Loss of Production during De-piling bogie changeover.	Countermeasure (Engineering solution):					Base Data 96 sheets/Day i.e. 6 M.T production is lost to recycling unit										
	Before :	After :	Target	Target - 0 Sheets Loss												
			Kaizen Start Kaizen Finish	16-10-2019 13-11-2019												
	Before condition : Previously for every De-piling bogie changeover, we are losing 1 sheet to recycling unit due to Template un-availability for piling					Team Members 1) Y. Kesavaraj - Production 2) M. Vijaya Kumar - Electrical 3) N. Karthikeyan - Production 4) Narayanasamy - Operator 5) Pillai - Fabricator Benefits: (P,Q,C,D,S,M) 1) P - Productivity improvement of 1900 M.T/Year 2) Q - Quality improved due to reduction in consumption of recycled waste. 3) C - Cost Saving of INR. 1.36 Cr/Annum & less Investment Cost of INR. 20,000/- due to in-house built. 4) D - Machine capacity has increased by 1900 M.T/Year 5) S - No Safety issues. 6) M - Morale increased due to productivity Sustenance: Training on SOP provided for operation, regular inspection & maintenance to										
Analysis why - why analysis :	Results :															
Problem Phenomena :	Green Sheets Production Loss / Day  Average Monthly Production 															
Loss of Production.																
Why 1 : Green sheets are sent to recycling																
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Root Cause : There are no templates available to supply for piling Green sheets in bogie.	Scope & Plan for Horizontal Deployment <table border="1"> <thead> <tr> <th>SN</th> <th>M/c</th> <th>Target date</th> <th>Responsibility</th> <th>Status</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>Line 2</td> <td>30-12-2019</td> <td>Maintenance HOD</td> <td>Completed</td> </tr> </tbody> </table>						SN	M/c	Target date	Responsibility	Status	1	Line 2	30-12-2019	Maintenance HOD	Completed
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Thank You